Initial Data Cleaning

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```
library(tidyverse)
library(readxl)
```

Loading the survey data

```
# Columns A-K hold the survey responses
# columns M onward hold summary statistics

# function loads FY26 survey into R

#opening Excel file at our Path, and reading only columns A:K as that's where our survey ansu

# We load everything as text so we are type safe.

read_fy26 <- function(path) {
  read_excel(path,
             sheet = "FY26",
             range = cell_cols("A:K"),
             col_types = "text")
}

# Opening two spreadsheets, loading FY26 survey into R

faculty <- read_fy26("Raw_Data/faculty_commute.xlsx")
staff   <- read_fy26("Raw_Data/staff_commute.xlsx")

nrow(faculty)  # should be 112
```

```
[1] 112
```

```
nrow(staff)    # should be 355
```

```
[1] 355
```

Combining and cleaning the workbooks

```
# name of code chunk

#| label: clean

# taking two tables and stacking them into one called commute

#bind_rows: stacking faculty and staff tables on top of one another, to create one table with

commute <- bind_rows(faculty, staff) |>


  # names are empty in FY26, and the real employee IDs aren't    needed
  # removing three columns and dropping FirstName and LastName to protect anonymity
  # names have space in them, so included backticks for EmployeeID
  select(-FirstName, -LastName, -`Employee ID`) |>

  # cleaning up columns with white space
  mutate(across(everything(), str_trim)) |>

  rename(zip = Zip,
         miles = `Miles To Work (one way)`,
         employee_type = `Employee Type`) |>

  # creating a new column numbering each person to create an anonymous ID
  mutate(person_id = row_number(),
  # so "staff" -> "Staff"
         employee_type = str_to_title(employee_type),
  # "91104-2946" -> "91104"
         zip = str_sub(zip, 1, 5),
  # converting miles from text back to numbers

         miles = as.numeric(miles)) |>
  # moving ID column to front of table
  relocate(person_id)
```

```r
count(commute, employee_type)  # count check
```

```
# A tibble: 2 x 2
  employee_type     n
  <chr>         <int>
1 Faculty         112
2 Staff           355
```

One row per person per day

```r
# overall objective: we need to reshape the data so that
# each row is one person on one day, and we know the meaning of each mode code

# decoder table: what each survey mode code means
mode_lookup <- tribble(
  ~code, ~mode,                    ~category,          ~day_type,     ~is_sov, ~occupants,
  "A",   "Zero Emission Vehicle",  "Electric Vehicle", "commute",     FALSE,   1,
  "B",   "Bus",                    "Public Bus",       "commute",     FALSE,   NA,
  "C",   "Rail / Plane",           "Commuter Rail",    "commute",     FALSE,   NA,
  "D",   "Walk",                   "Walk",             "commute",     FALSE,   NA,
  "E",   "Bicycle",                "Bike",             "commute",     FALSE,   NA,
  "F",   "Telecommute",            "Telecommuting",    "telecommute", FALSE,   NA,
  "G",   "Noncommuting",           "No trip",          "no_trip",     FALSE,   NA,
  "H",   "Drive Alone",            "Automobile",       "commute",     TRUE,    1,
  "I",   "Motorcycle",             "Automobile",       "commute",     TRUE,    1,
  "J",   "2 persons in vehicle",   "Carpool",          "commute",     FALSE,   2,
  "K",   "3 persons in vehicle",   "Carpool",          "commute",     FALSE,   3,
  "L",   "4 persons in vehicle",   "Carpool",          "commute",     FALSE,   4,
  "M",   "5 persons in vehicle",   "Carpool",          "commute",     FALSE,   5,
  "N",   "6 persons in vehicle",   "Carpool",          "commute",     FALSE,   6,
  "O",   "7 persons in vehicle",   "Carpool",          "commute",     FALSE,   7,
  "P",   "8 persons in vehicle",   "Carpool",          "commute",     FALSE,   8,
  "Q",   "9 persons in vehicle",   "Carpool",          "commute",     FALSE,   9,
  "R",   "10 persons in vehicle",  "Carpool",          "commute",     FALSE,   10,
  "S",   "11 persons in vehicle",  "Carpool",          "commute",     FALSE,   11,
  "T",   "12 persons in vehicle",  "Carpool",          "commute",     FALSE,   12,
  "U",   "13 persons in vehicle",  "Carpool",          "commute",     FALSE,   13,
  "V",   "14 persons in vehicle",  "Carpool",          "commute",     FALSE,   14,
  "W",   "15 persons in vehicle",  "Carpool",          "commute",     FALSE,   15,
```

```
  "X",   "3/36 Work Week Days Off", "No trip",          "no_trip",     FALSE,   NA,
  "Y",   "4/40 Work Week Day Off",  "No trip",          "no_trip",     FALSE,   NA,
  "Z",   "9/80 Work Week Day Off",  "No trip",          "no_trip",     FALSE,   NA,
  "AA",  "Vacation",                "No trip",          "no_trip",     FALSE,   NA,
  "BB",  "Sick",                    "No trip",          "no_trip",     FALSE,   NA,
  "CC",  "Unknown (ask Alexis)",    "Unknown",          "commute",     FALSE,   NA
)

# before this code, each person's row of five days was five separate rows.

# now, we'll have one row per person per day, and we'll have the meaning attached

commute_days <- commute |>
  # turns five day columns into five rows
  pivot_longer(MonMode:FriMode, names_to = "day", values_to = "code") |>

  # changes "MonMode" to "Mon"
  mutate(day = str_remove(day, "Mode")) |>

  # looks up each code in decoder table, adds meaning as new columns
  left_join(mode_lookup, by = "code")

nrow(commute_days)                  # should be 2,335 (467 people x 5 days)
```

```
[1] 2335
```

```
commute_days |> filter(is.na(mode)) # should be 0 rows (every code matched)
```

```
# A tibble: 0 x 11
# i 11 variables: person_id <int>, zip <chr>, miles <dbl>, employee_type <chr>,
#   day <chr>, code <chr>, mode <chr>, category <chr>, day_type <chr>,
#   is_sov <lgl>, occupants <dbl>
```

```
# cleaning process where we remove days where nobody traveled, and then we calculate what sha

# doing this separately for faculty and staff

#| label: trips

# drop days with no trip (vacation, sick, days off, noncommuting)
```

```r
trips <- commute_days |>

  # by filtering, we're able to only keep the rows that have day_type not being no trip (so w
  filter(day_type != "no_trip")

# we count the rows for each combination
count(trips, employee_type, category) |>

  # telling R next step should be done separately for faculty and staff
  group_by(employee_type) |>

  # dividing each count by group's total so we can get share of days and round to 3 clos
  mutate(share = round(n / sum(n), 3))
```

```
# A tibble: 16 x 4
# Groups:   employee_type [2]
   employee_type category            n share
   <chr>         <chr>           <int> <dbl>
 1 Faculty       Automobile        160 0.297
 2 Faculty       Bike               74 0.137
 3 Faculty       Carpool            23 0.043
 4 Faculty       Commuter Rail       7 0.013
 5 Faculty       Electric Vehicle   71 0.132
 6 Faculty       Telecommuting      56 0.104
 7 Faculty       Unknown            36 0.067
 8 Faculty       Walk              112 0.208
 9 Staff         Automobile        984 0.585
10 Staff         Bike               65 0.039
11 Staff         Carpool           104 0.062
12 Staff         Commuter Rail      40 0.024
13 Staff         Electric Vehicle  176 0.105
14 Staff         Telecommuting     171 0.102
15 Staff         Unknown            40 0.024
16 Staff         Walk              102 0.061
```

Checking against the Sustainability Annual Report

```r
#| label: check-within-5

#checking cleaned data against the college's published stats
```

```r
# the 2026 SAR says 72% of faculty and 43% of staff are 5 miles within campus
#cross-referencing to ensure cleaning is correct

# SAR: 72% of faculty and 43% of staff live within 5 miles
commute |>

# group_by makes next step run separately for faculty and staff
  group_by(employee_type) |>

# collapsing each group into one row of results, so that we know how many people are in each

  summarize(respondents = n(),
            pct_within_5_miles = round(100 * mean(miles <= 5), 1))
```

```
# A tibble: 2 x 3
  employee_type respondents pct_within_5_miles
  <chr>               <int>              <dbl>
1 Faculty               112               72.3
2 Staff                 355               42.8
```

```r
# here, we're calculating the weekly SOV miles from data and comparing them to the numbers i

# SOV miles are necessary because the 2030 goal is a 30% cut in SOV miles

trips |>

  # we're keeping only the SOV days, so we are making sure we're keeping the rows where it's
  filter(is_sov) |>

  # group_by runs the next step separately based on whether respondent is a faculty or staff
  group_by(employee_type) |>

  # adding up miles based on the days we've selected
  # each row is one day's one-way distance, and the total is SOV miles for one survey week
  summarize(sov_miles_per_week = sum(miles))
```

```
# A tibble: 2 x 2
  employee_type sov_miles_per_week
  <chr>                      <dbl>
1 Faculty                     1884
2 Staff                      12735
```

open question: what does CC mean?